

Breakdown of the Users Stories into tasks and prioritization of them.

Must Have:

Admin Stories

As an admin, I want to add, delete and modify start-ups information so that I can update the Village's information and make the start-up visible on the website.

Tasks :

- Adding a Startup entity to the database.
- Adding a Startup repository file to manage the CRUD operations on the startup table.
- Adding a Startup class file to validate the format of the data that we will store in our database.
- Adding a Startup Services file to manage the business logic of the startups.
- Adding a Startup controller file that will call the appropriate functions as needed for the route.
- Adding a Startup route file that will include all the routes the admin need to realise these operations and will call all the needed controllers.
- Adding a startups space page for the admin so they can modify their informations.

As an admin, I want to add, delete and modify types associated with my start-ups so that I can update the start-ups information and this way visitors can sort startups according to their type on the website.

Tasks :

- Adding a Type entity to the database.
- Adding a Type repository file to manage the CRUD operations on the startup table.
- Adding a Type class file to validate the format of the data that we will store in our database.
- Adding a Type Services file to manage the business logic of the types.
- Adding a Type controller file that will call the appropriate functions as needed for the route.
- Adding a Type route file that will include all the routes the admin need to realise these operations and will call all the needed controllers.
- Adding a types space page for the admin so they can modify their informations.

As an admin, I want to post, update and delete blog articles, so that I can make the website more attractive and increase the number of visits.

Tasks :

- Adding a Post entity to the database.
- Adding a Post repository file to manage the CRUD operations on the startup table.
- Adding a Post class file to validate the format of the data that we will store in our database.
- Adding a Post Services file to manage the business logic of the posts.
- Adding a Post controller file that will call the appropriate functions as needed for the route.
- Adding a Post route file that will include all the routes the admin need to realise these operations and will call all the needed controllers.
- Adding a posts space page for the admin so they can modify their informations.

As an admin, I want to add, update and delete the partners lists and information, so that I can keep the website updated with new partnerships and make them visible on the website.

Tasks :

- Adding a Partner entity to the database.
- Adding a Partner repository file to manage the CRUD operations on the startup table.
- Adding a Partner class file to validate the format of the data that we will store in our database.
- Adding a Partner Services file to manage the business logic of the partners.
- Adding a Partner controller file that will call the appropriate functions as needed for the route.
- Adding a Partner route file that will include all the routes the admin need to realise these operations and will call all the needed controllers.
- Adding a partners space page for the admin so they can modify their informations.

For all admin routes :

- Adding a function that generate a token with the role of the user so only the admin can do these verifications.
- Adding a function that will read the token and verify the role of the user to forbid other people except the admin to access these routes.

For all Entities that need to use pictures :

- Create a Picture entity in the database
- Configure Cloudinary to host the picture so we can manipulate them.
- Create a Cloudinary function that upload pictures to Cloudinary
- Create a multerv middleware that intercept upload files

Start-up Stories

As a Start-up, I want to update my account information, so that I can keep my information updated for the visitors.

Tasks :

- Adding a User entity to the database.
 - Adding a User repository file to manage the CRUD operations on the startup table.
 - Adding a User class file to validate the format of the data that we will store in our database.
 - Adding a User Services file to manage the business logic of the users.
 - Adding a User controller file that will call the appropriate functions as needed for the route.
 - Adding a User route file that will include all the routes the logged-in user need to realise these operations and will call all the needed controllers.
 - Adding a Startup route file that will include all the routes the logged-in user need to realise these operations and will call all the needed controllers.
 - Adding a function that will read the token and verify the role of the user to forbid other people except the admin and the logged-in user to access these routes.
- Adding a User route file that will include all the routes the admin need to realise these operations and will call all the needed controllers.

As a Start-up I want to know what are the partners of the association, so I can learn what are the association's network

- Adding a partners page for the visitors so they can see the partners informations.

As a Start-up I want to write a quote that describe my experience with the association so that I can inform other start-up about the quality of the association.

Tasks :

- Adding a Quote entity to the database.
- Adding a Quote repository file to manage the CRUD operations on the startup table.
- Adding a Quote class file to validate the format of the data that we will store in our database.
- Adding a Quote Services file to manage the business logic of the quotes.
- Adding a Quote controller file that will call the appropriate functions as needed for the route.
- Adding a Quote route file that will include all the routes the logged-in user need to realise these operations and will call all the needed controllers.

- Adding a Quote route file that will include all the routes the admin need to realise these operations and will call all the needed controllers.

Visitor Stories

As a visitor, I want to take a rendez-vous to know more about how the Village and the Mayenne environment can help me in my current situation, so that I know who I can contact.

- Adding a Contact transporter that will send email to the director of the association with the message of the visitor.
- Adding a Contact class file that will validate the data that want to be send to the director of the association.
- Adding a Contact services file that will contain the business logic of the contact form and use the transporter to send the email.
- Adding a Contact controller file that will use the adequate function in the Contact services.
- Adding a Contact route that everybody can access to contact the association.
- Adding a Contact form page for the visitor to use to send their email.
- Adding a field in the subject of the contact form that said « I want to know more about the Mayenne environment ».

As a visitor, I want to know what are the upcoming events in the Village, so that I can take part in events that interest me.

- Adding an Agenda page that will display all the informations about the upcoming events, and the news of the association.

As a visitor, I want to discover the start-ups present in the Mayenne department so that I can contact them for a collaboration if one of them is interesting to me.

-Adding a Startups page that display all the start-ups and their informations notably their website and email so they can discover and contact them.

As a visitor, I want to know what services the Village by CA is delivering so that I can know if the Village by CA can respond to my needs.

-Adding a field in the subject of the contact form that said « I want to know more about how the association can respond to my needs ».

Should Have:

Admin stories:

As an admin, I want to add, update and delete room for rent information, so that I can keep the visitors updated with available rooms.

- Adding a Location entity to the database.
- Adding a Location repository file to manage the CRUD operations on the startup table.
- Adding a Location class file to validate the format of the data that we will store in our database.
- Adding a Location Services file to manage the business logic of the locations.
- Adding a Location controller file that will call the appropriate functions as needed for the route.
- Adding a Location route file that will include all the routes the admin need to realise these operations and will call all the needed controllers.
- Adding a locations space page for the admin so they can modify their informations.

As an admin, I want to create, delete and update new events on an agenda, so that I can keep start-ups and visitors informed of the upcoming events in the Village.

- Adding a Event entity to the database.
- Adding a Event repository file to manage the CRUD operations on the startup table.
- Adding a Event class file to validate the format of the data that we will store in our database.
- Adding a Event Services file to manage the business logic of the events.
- Adding a Event controller file that will call the appropriate functions as needed for the route.
- Adding a Event route file that will include all the routes the admin need to realise these operations and will call all the needed controllers.
- Adding a events space page for the admin so they can modify their informations.

Start-up Story :

As a Start-up, I want to see the upcoming events in the Village agenda, so that I can plan to come to the events that interest me.

-Adding an Agenda page that will display all the informations about the upcoming events, and the news of the association.

Visitor Story:

As a visitor, I want to rent a place so that I can organize some events here.

-Adding a location page where I can see all the locations and their details and a field in the subject of the contact form that said « I want to rent a place ».

As a visitor, I want to have a partnership with the Village by Ca so that I can help them in their activities and collaborate with them.

-Adding a field in the subject of the contact form that said « I want to become a partner of the Village by CA ».

As a visitor, I want to contact the Village headmaster, so that I can ask various questions about the events or the Village.

-Adding a field in the subject of the contact form that said « I want to know more about the Village by CA ».

Could Have:

Visitor and Start-up Story:

As a visitor I want to talk with a chatbot so that I have my answer quickly and don't bother the administrator with very simple questions.

-Configure a basic Chatbot.

Won't Have:

Start-up Story:

As a start-up, I want to have a list of the actors in the Mayenne ecosystem so that I know who can help my business.

-Adding a page with all the actors of the Mayenne ecosystem.

Dependencies between tasks :

1 : Create the database in order to be able to use the repo files that manipulate the data in the database. → Maï

2 : Create the repo files in order to use the services files so we know if the business logic is applied well. → Maï

3 : Create the configuration of Cloudinary so the services can upload into Cloudinary and save the picture data in the database so we can insert entities that use pictures. → Maï

4 : Create the services files so the controllers have functions to call, even if we can see if the controller is correctly working with just a simple response. → Guillaume and Maï

5 : Create the controllers files so the routes can return something via controllers. → Guillaume

6 : Create a login controller that generate JWT in order to use middlewares that verify the user role and the token presence. → Maï

7 : Finishing all the backend in order to display the data to the frontend. → Guillaume and Maï

Sprint Planning Overview

Here is the sprint plan used to create the Village by CA application, followed by detailed descriptions of each of them :

- Idea Developement
- Project Planning
- Technical Documentation
- Mockups Creation and Client Alignment
- Technical Knowledge Acquisition
- Database Setup and Basic Backend Foundation
- Repository Structure and Backend Completion
- Services Layer Implementation
- Frontend Technology Familiarization
- Frontend Progression and Bug Fixing

Sprints Details :

Idea Development

Sprint Duration : 1 Week

Deadline : Sunday, Dec 7

Initial exploration of the concepts, identification of core objectives, and clarification of expected outcomes.

- Developers explore initial concepts and define the core product vision.
 - Brainstorming several ideas : Maï and Guillaume
 - Ideas evaluation : Maï and Guillaume
 - Decision and refinement : Maï and Guillaume
- SCM ensures early project structure and repository readiness.
- QA reviews feasibility aspects and identifies potential risks or constraints

Task completion : 3 / 3

Sprint velocity : 100%

Scrum Master : Maï Le Meur

What worked well during the sprint

- Strong collaboration during brainstorming and idea evaluation.
- Clear definition of the project vision and objectives.
- Early SCM setup provided a solid repository foundation.
- QA identified feasibility constraints early.
- All tasks were completed successfully.

What challenges did we face

- Initial ambiguity around the scope required extra clarification.
- Aligning conceptual directions took additional discussion time.

What changes can we make to improve the next sprint

- Document early decisions more systematically.

Project Planning

Sprint Duration : 1 Week

Deadline : Sunday, Dec 14

Definition of scope, task breakdown, prioritization, and sprint structuring to ensure a clear development roadmap.

- Developers break down features into actionable tasks and estimate workload.
 - Creation of a Gant chart : Maï and Guillaume
- SCM prepares branching strategies and repository conventions.
- QA defines the initial testing strategy and acceptance criteria.

Task completion : 2 / 2

Sprint velocity : 100%

Scrum Master : Guillaume Leray Girardeau

What worked well during the sprint

- Efficient breakdown of features into actionable tasks.
- Smooth collaboration on the Gantt chart.
- Full task completion with strong coordination.

What challenges did we face

- Estimating workload for unfamiliar features required extra effort.
- Aligning priorities across roles took time.

What changes can we make to improve the next sprint

- Introduce estimation guidelines for consistency.
- Hold a short alignment meeting before finalizing sprint scope.

Technical Documentation

Sprint Duration : 2 Weeks

Deadline : Sunday, Jan 11

Creation of foundational documentation, including architecture notes, workflow descriptions, and technical specifications.

- Developers produce architectural diagrams, workflow descriptions, and technical specifications.
 - Define User Stories and Mockups : Guillaume and Maï
 - Complete Figma Training : Guillaume
 - Design System Architecture : Maï
 - Define Components, Classes and Database Design : Maï
 - Create High-level Sequence Diagram : Maï
 - Document Internal and External API : Maï
 - Plan SCM and QA Strategies : Guillaume
- SCM validates documentation consistency with the planned development workflow.
- QA reviews documentation to ensure testability and clarity of requirements.

Task completion : 6 / 6

Sprint velocity : 100%

Scrum Master : Maï Le Meur

What worked well during the sprint

- Comprehensive documentation across architecture, workflows, APIs, and design.
- Clear division of responsibilities between team members.
- SCM and QA ensured consistency and testability.
- All tasks were completed with high quality.

What challenges did we face

- The large documentation volume required significant coordination.
- Some technical decisions evolved mid-sprint and required updates.

What changes can we make to improve the next sprint

- Adopt documentation versioning to track changes.
- Schedule mid-sprint syncs to maintain alignment.

Mockup Creation & Client Alignment

Sprint Duration : 1 Week

Deadline : Sunday, Jan 18

Design of interface mockups using Figma and validation with the client (Amandine Chemin) to ensure the proposed solution meets expectations.

- Developers design UI mockups and refine them based on feedback.
 - Create visual mockups using Figma : Maï and Guillaume
 - Hold a meeting with Amandine Chemin to establish a clear design direction : Maï and Guillaume
- SCM stores and versions design assets in the repository.
- QA verifies that mockups meet usability expectations and match client requirements validated with Amandine Chemin.

Task completion : 2 / 2

Sprint velocity : 100%

Scrum Master : Guillaume Leray Girardeau

What worked well during the sprint

- Effective creation of mockups using Figma.
- Productive meeting with the client that clarified design expectations.

What challenges did we face

- Some design elements required multiple iterations.
- Limited time to explore alternative design options.

What changes can we make to improve the next sprint

- Gather preliminary client input earlier.
- Allocate buffer time for design iterations.

Technical Knowledge Acquisition

Sprint Duration : 2 Weeks

Deadline : Sunday, Feb 1

Learning and familiarization with required technologies such as Express, Node.js, Prisma, and Docker through official documentation.

- Developers study Express, Node.js, Prisma, and Docker through official documentation.
 - Read documentation and complete a training course on Express and [Node.js](#) : Guillaume
 - Read documentation on Prisma : Maï
 - Read documentation on Docker : Maï
- SCM provides environment setup guidelines and ensures tools are correctly configured.
- QA prepares early test scenarios based on the technologies being adopted.

Task completion : 3 / 4

Sprint velocity : 75%

Scrum Master : Maï Le Meur

What worked well during the sprint

- Strong engagement with documentation and training materials.
- Clear division of learning topics.

What challenges did we face

- One task remained incomplete due to time constraints.
- Some technologies required more learning time than expected.

What changes can we make to improve the next sprint

- Allocate more time for deep technical learning earlier.
- Use pair-learning sessions to accelerate understanding.

Database Setup & Backend Foundations

Sprint Duration : 1 Week

Deadline : Sunday, Feb 8

Creation of the database schema and implementation of initial backend components, including basic routes, controllers, and class structures.

- Developers create the database schema and implement initial routes, controllers, and class structures.
 - Creation of basic API routes : Guillaume
 - Implementation of basic controllers : Guillaume
 - Initialization of the Database : Maï
 - Development of the core class structure : Maï
- SCM reviews early backend code and validates repository organization.
- QA tests basic endpoints using Postman and verifies database interactions.

Task completion : 3 / 4

Sprint velocity : 75%

Bug count : 1

Resolution rate : 100%

Scrum Master : Guillaume Leray Girardeau

What worked well during the sprint

- Successful creation of the database schema and backend structure.
- Good collaboration on routes, controllers, and classes.
- SCM validated repository organization.
- QA effectively tested endpoints and database interactions.

What challenges did we face

- One task remained incomplete.
- A bug was identified but resolved quickly.

What changes can we make to improve the next sprint

- Improve time estimation for backend setup tasks.
- Introduce early code reviews to catch issues sooner.

Repository Structure & Backend Completion

Sprint Duration : 1 Week

Deadline : Sunday, Feb 15

Setup of repository files and completion of backend classes, controllers, and structural elements.

- Developers set up the repository files and complete the initial backend components, including classes, controllers, and core structural elements.
 - Final classes creation : Maï
 - Finishing controllers implementation : Guillaume
- SCM reviews the repository organization and validates backend code structure before merging into the development branch.
- QA performs early functional checks to ensure that backend routes, controllers, and structural elements behave as expected.

Task completion : 3 / 5

Sprint velocity : 60%

Bug count : 8

Resolution rate : 100%

Scrum Master : Maï Le Meur

What worked well during the sprint

- Repository structure was properly set up and validated.
- Backend classes and controllers progressed significantly.
- QA performed early functional checks.
- All identified bugs were resolved.

What challenges did we face

- Only 60% of tasks were completed due to complexity.

- High bug count indicated structural issues.

What changes can we make to improve the next sprint

- Focus on backend consistency before expanding features.
- Implement stricter coding standards to reduce bugs.

Service Layer Implementation

Sprint Duration : 1 Week

Deadline : Sunday, Feb 22

Development of service modules and integration with existing controller logic to support business operations.

- Developers implement service modules and connect them to controllers.
 - Services file creation : Maï and Guillaume
 - Implementation of prisma transaction : Guillaume
 - Connecting services to controllers : Maï
- SCM reviews service logic and validates integration with existing backend components.
- QA tests service endpoints and verifies business logic correctness

Task completion : 4 / 5

Sprint velocity : 80%

Bug count : 33

Resolution rate : 85%

Scrum Master : Guillaume Leray Girardeau

What worked well during the sprint

- Service modules were successfully created and integrated.
- Prisma transactions were implemented effectively.
- SCM validated service logic.
- QA tested business logic thoroughly.

What challenges did we face

- One task remained incomplete.
- High bug count (33) indicated complexity in service logic.
- Some integration points required rework.

What changes can we make to improve the next sprint

- Improve service design before implementation.
- Increase pair programming to reduce inconsistencies.

Frontend Technology Familiarization

Sprint Duration : 1 Week

Deadline : Sunday, Mar 1

Introduction to the frontend stack, implementation of Cloudinary and JWT, creation of authentication middlewares, and development of the contact form.

- Developers explore the frontend stack with React and Vite and implement Cloudinary, JWT, authentication middlewares, and the contact form.
 - Read documentation and complete a training course on React and Vite : Guillaume
 - Create basic frontend routes and architecture : Guillaume
 - Read documentation about Cloudinary : Maï
 - Implement authentication with JWT : Maï
 - Add contact form to the frontend : Maï
- SCM ensures proper frontend folder structure and asset management.
- QA tests authentication flows, media uploads, and form submissions

Task completion : 3 / 4

Sprint velocity : 75%

Bug count : 21

Resolution rate : 95%

Scrum Master : Maï Le Meur

What worked well during the sprint

- Strong progress in learning React, Vite, Cloudinary, and JWT.
- Successful implementation of authentication and the contact form.
- SCM ensured a clean frontend structure.
- QA validated authentication flows and media uploads.

What challenges did we face

- One task remained incomplete.
- Integration of new technologies required extra debugging.

What changes can we make to improve the next sprint

- Provide more time for frontend onboarding.
- Create small prototype components before full integration.

Frontend Progression & Bug Fixing

Sprint Duration : 1 Week

Deadline : Sunday, Mar 8

Advancement of frontend features, implementation of formatting utilities, and correction of identified bugs.

- Developers advance UI implementation, add formatting functions, and fix identified bugs.
 - Connect contact form to the backend : Maï
 - Creating React components : Guillaume
 - Adding missing frontend pages with new components : Maï
 - Creation of a test suite with Jest : Maï
- SCM reviews frontend code and ensures consistency with coding standards.
- QA validates UI behavior, checks formatting logic, and confirms bug resolutions

Task completion : 3 / 4

Sprint velocity : 75%

Bug count : 7

Resolution rate : 85%

Scrum Master : Guillaume Leray Girardeau

What worked well during the sprint

- Significant progress on UI components and missing pages.
- Contact form successfully connected to the backend.
- Jest test suite creation improved testing capabilities.
- QA validated UI behavior and bug fixes.

What challenges did we face

- One task remained incomplete.
- Some bugs required multiple iterations.

What changes can we make to improve the next sprint

- Increase automated testing coverage to reduce regressions.